BINGHAMTON UNIVERSITY
DEPARTMENT OF MATHEMATICAL SCIENCES

Math 147

Elementary Statistics

Course notes, with every exercise worked out

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About these notes

These are the notes I use in the classroom for Math 147, Elementary Statistics, rewritten so that they can be read away from the blackboard. Nothing has been added for the sake of completeness and nothing has been smoothed over: the order of the topics, the examples and the exercises are the ones we actually do in class.

What is new here is that *every exercise is worked out*. In the original handwritten pages many problems were left for the student, and several solutions were sketched in two lines because I was talking while I wrote them. Here each one is carried out step by step, with the reasoning that justifies the step and not only the arithmetic.

Two conventions are worth stating before we start. A box marked REMEMBER holds something you will use again and again, and is worth memorising. A box marked CAREFUL marks a place where students reliably go wrong; I have put one wherever I have seen the same mistake more than twice.

> **HOW TO USE THIS**
>
> Read the statement, close the book, and try the exercise. Only then read the solution. A solution you have read is not a solution you can produce; statistics is learned with a pencil in your hand.

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CHAPTER 1

Getting Ready for Statistics

Statistics is not an arithmetic course, but it does run on arithmetic. Before we meet a single distribution we need to be fluent with four things: proportions and percentages, rounding, reading a formula, and summation notation. This chapter is that warm-up. Every exercise from the first handout appears here, fully worked.

PROPORTION AND PERCENTAGE

A **proportion** is a fraction $\frac{\#1}{\#2}$, where #2 is the whole (the larger number, the total) and #1 is the part of it you care about. It always lands between 0 and 1. A **percentage** is the same number with the decimal point moved two places to the right:

$$\text{percentage} \;=\; \text{proportion} \times 100\%.$$

The whole difficulty in these problems is deciding what plays the role of #2. Decide that first, and the arithmetic follows.

1.1 Proportions and percentages

Exercise 1

Sage pays \$550 per month for rent and her monthly salary is \$1876.

(a) What proportion of her monthly salary is her rent? (This answer will be between 0.000 and 1.000.)

(b) What percentage of her monthly salary is her rent? (This answer will be between 0.0% and 100.0%.)

Solution. The whole here is the salary, because the rent is a part of it. So #2 $= 1876$ and #1 $= 550$.

(a) The proportion is

$$\frac{550}{1876} = 0.29317\ldots \approx 0.293.$$

(b) Multiplying by 100,

$$0.29317\ldots \times 100 = 29.317\ldots\,\% \approx 29.32\%.$$

Rent eats a little under a third of what she earns.

(a) 0.293 (b) 29.32%

Exercise 2

At Binghamton High School there are 295 seniors who entered Binghamton schools in the fifth grade or lower, and 798 seniors who entered in the sixth grade or higher. What percentage of the seniors entered Binghamton schools in the fifth grade or lower?

Solution. The whole is *all* the seniors, and that number is not given directly: we have to build it.

$$\#2 = 295 + 798 = 1093, \qquad \#1 = 295.$$

Then

$$\frac{295}{1093} = 0.26989\ldots \approx 26.99\%.$$

26.99%

Exercise 3

In a particular arts and humanities college, the following table gives the numbers of men and women in various majors.

	Men	**Women**	**Total**
Anthropology	83	69	152
History	102	110	212
Music	53	37	90
Total	238	216	454

(a) What percentage of the women are anthropology majors?

(b) What percentage of the music majors are men?

Solution. Read each question backwards: the group named *after* the word "of" is the whole.

(a) "of the women" ⇒ the whole is all 216 women, and the part is the 69 women in anthropology:

$$\frac{69}{216} = 0.31944\ldots \approx 31.94\%.$$

(b) "of the music majors" ⇒ the whole is the 90 music majors, and the part is the 53 men among them:

$$\frac{53}{90} = 0.58888\ldots \approx 58.89\%.$$

(a) 31.94% (b) 58.89%

CAREFUL

Notice how different (a) and (b) are, even though both use numbers from the same table. $\frac{69}{216}$ and $\frac{69}{152}$ answer two different questions: "what percentage of the women study anthropology" and "what percentage of the anthropology students are women".

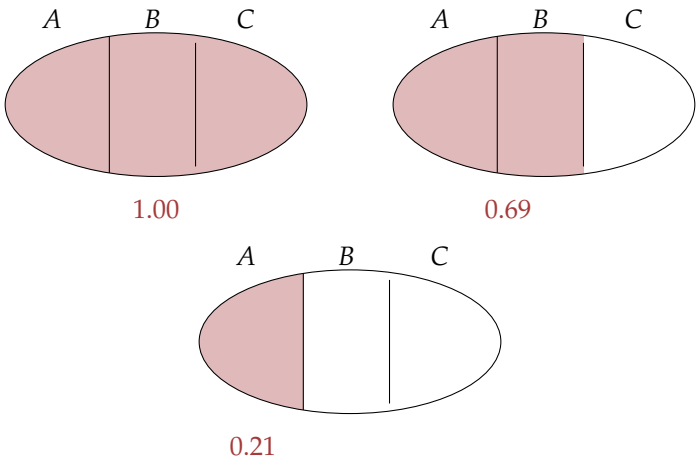
> Choosing the wrong denominator is the single most common mistake in this chapter.

1.2 Areas as proportions

Proportions do not have to come from counting people. An area can be a proportion too, and this is exactly how we will read probability later in the course.

Exercise 4

A figure is divided into three non-equal pieces by lines. The three pictures below are all of the same figure with the same division lines. Different areas are shaded and labelled on each. Find the areas of the portions labelled B and C.



Solution. Name the three unknown areas A, B, C. Each picture gives one equation:

$$A + B + C = 1.00, \qquad A + B = 0.69, \qquad A = 0.21.$$

Now subtract, from the smallest piece of information upwards. Taking the first two,

$$C = (A + B + C) - (A + B) = 1.00 - 0.69 = 0.31,$$

and taking the last two,

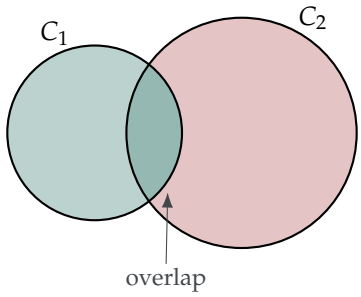
$$B = (A + B) - A = 0.69 - 0.21 = 0.48.$$

As a check, $0.21 + 0.48 + 0.31 = 1.00$, as it must be.

$B = 0.48, \qquad C = 0.31$

Exercise 5

Two circles overlap as sketched below. The area of circle 1 is 0.48 and the area of circle 2 is 0.57.



(a) Can you figure out the shaded area or do you need more information? What would that be?

(b) The overlapping area of the two circles is 0.16. Can you now find the shaded area?

Solution. **(a)** No, not yet. In Exercise 4 the pieces did not overlap, so areas simply added up. Here the two circles share a region, and if we write $0.48 + 0.57$ that shared region has been counted *twice*. What is missing is exactly the area of the overlap.

(b) With the overlap equal to 0.16, count circle 1 without its shared part, and then all of circle 2:

$$\text{shaded} = \underbrace{(0.48 - 0.16)}_{\text{only } C_1} + \underbrace{0.57}_{C_2} = 0.32 + 0.57 = 0.89.$$

Equivalently, add everything and remove the double count: $0.48 + 0.57 - 0.16 = 0.89$.

0.89

OVERLAPPING PIECES

When two regions overlap,

$$\text{area}(C_1 \text{ or } C_2) = \text{area}(C_1) + \text{area}(C_2) - \text{area}(C_1 \text{ and } C_2).$$

This is the addition rule, and we will meet it again word for word when we do probability.

1.3 Rounding and the calculator

HOW TO ROUND

To round to a given decimal place, look at the *next* digit to the right. If it is 5 or more, add one to the digit you are keeping; if it is 4 or less, leave it alone. Then drop everything after it.

Exercise 6

Round 57.39283 to the nearest thousandth.

Solution. "Thousandth" means three decimal places, so we keep 57.392 and look at the fourth decimal:

$$57.392\,\underline{8}3.$$

That digit is $8 \geq 5$, so the third decimal goes up from 2 to 3.

57.393

Exercise 7

Round 0.02397 to the nearest hundredth.

Solution. "Hundredth" means two decimal places, so we keep 0.02 and look at the third decimal:

$$0.02\,\underline{3}97.$$

That digit is $3 \leq 4$, so the second decimal stays as it is. The 97 that follows is irrelevant — only the first discarded digit decides.

0.02

CAREFUL

In Exercise 7 it is tempting to look at the 9 and round up, reasoning that 0.02397 is "almost 0.024". Rounding is decided by one digit only, the first one you drop. Chaining the rounding ($0.02397 \to 0.0240 \to 0.024 \to 0.02$) can give a different answer and is wrong.

Exercise 8

Use your calculator to find $\sqrt{21}$, correct to 3 decimal places.

Solution. The calculator gives $\sqrt{21} = 4.58257569\ldots$. Keeping three decimals we look at the fourth, which is 5, so the third goes up from 2 to 3.

$\sqrt{21} \approx 4.583$

Exercise 9

Use your calculator to find 18^2.

Solution. Squaring means multiplying the number by itself: $18^2 = 18 \times 18 = 324$.

324

Exercise 10

When a scientific calculator shows the answer for a computation as 5.38E − 04, what number does that represent?

Solution. The display 5.38E − 04 is the calculator's way of writing scientific notation:

$$5.38\text{E} - 04 \;=\; 5.38 \times 10^{-4}.$$

A negative exponent moves the decimal point to the *left*, as many places as the exponent says — here four places:

$$5.38 \to 0.538 \to 0.0538 \to 0.00538 \to 0.000538.$$

$5.38\text{E}-04 = 0.000538$

1.4 Inequalities and interval notation

Exercise 11

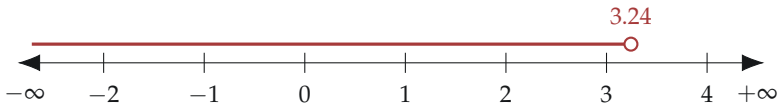
Consider the mathematical statement $x < 3.24$.

(a) How would you read that statement in words?

(b) Give the interval notation for all the numbers that make that statement true. Draw it on a number line.

Solution. **(a)** "x is less than 3.24."

(b) Every number strictly to the left of 3.24 works, and 3.24 itself does not. In interval notation that is $(-\infty, 3.24)$. The round bracket, and the hollow dot in the picture, both say "this endpoint is not included".



$(-\infty, 3.24)$

Exercise 12

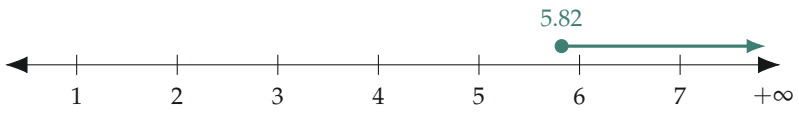
Consider the mathematical statement $y \geq 5.82$.

(a) How would you read that statement in words?

(b) Give the interval notation for all the numbers that make that statement true. Draw it on a number line.

Solution. **(a)** "y is greater than or equal to 5.82."

(b) Now 5.82 *is* allowed, because of the "or equal to". The square bracket, and the solid dot, both say "this endpoint is included".



$[5.82, +\infty)$

BRACKET OR PARENTHESIS

Use a square bracket [ ] when the endpoint belongs to the set ($\leq$ or $\geq$), and a round parenthesis ( ) when it does not ($<$ or $>$). Infinity never belongs to anything, so it always takes a parenthesis.

1.5 Solving for a letter

UNDOING OPERATIONS

To isolate a letter, undo what is being done to it, in reverse order. Each operation is undone by its opposite:

$$+ \longleftrightarrow -, \qquad \times \longleftrightarrow \div, \qquad (\,)^2 \longleftrightarrow \sqrt{\ }.$$

Whatever you do to one side of the equals sign, you do to the other.

Exercise 13

Solve for x: $5 + 3x = 18$.

Solution. The 5 is added, so subtract it from both sides; the 3 multiplies, so divide by it.

$$5 + 3x = 18 \implies 3x = 18 - 5 = 13 \implies x = \frac{13}{3}.$$

As a decimal, $x = 4.333\ldots \approx 4.333$. Check: $5 + 3 \cdot \frac{13}{3} = 5 + 13 = 18$. ✓

$x = \frac{13}{3} \approx 4.333$

Exercise 14

Solve for y: $7y - 8 = 24 + 26y$.

Solution. The letter appears on both sides, so first gather the y's on one side and the numbers on the other.

$$7y - 8 = 24 + 26y \implies 7y - 26y = 24 + 8 \implies -19y = 32 \implies y = -\frac{32}{19}.$$

As a decimal, $y = -1.6842\ldots \approx -1.684$. Check: $7(-1.6842) - 8 = -19.789$ and $24 + 26(-1.6842) = -19.789$. ✓

$y = -\frac{32}{19} \approx -1.684$

Exercise 15

Solve for r: $\dfrac{7}{5} = \dfrac{6}{r}$.

Solution. Two fractions set equal to each other: cross-multiply.

$$\frac{7}{5} = \frac{6}{r} \implies 7r = 5 \cdot 6 = 30 \implies r = \frac{30}{7}.$$

As a decimal, $r = 4.2857\ldots \approx 4.286$.

$r = \frac{30}{7} \approx 4.286$

Exercise 16

Solve for m: $\quad 3.9 = \dfrac{34.2}{m}$.

Solution. Here m sits in a denominator, so first bring it up by multiplying both sides by m:

$$3.9 = \frac{34.2}{m} \implies 3.9\,m = 34.2 \implies m = \frac{34.2}{3.9} = 8.7692\ldots$$

$m = \frac{34.2}{3.9} \approx 8.769$

Exercise 17

Solve for k: $\quad 9.7 = \dfrac{157.4}{\sqrt{k}}$.

Solution. Two steps, in this order: clear the denominator, then undo the square root by squaring.

$$9.7 = \frac{157.4}{\sqrt{k}} \implies 9.7\sqrt{k} = 157.4 \implies \sqrt{k} = \frac{157.4}{9.7} = 16.2268\ldots$$

Now square both sides, using $\left(\sqrt{k}\right)^2 = k$:

$$k = \left(\frac{157.4}{9.7}\right)^2 = (16.2268\ldots)^2 = 263.309\ldots$$

$k = \left(\frac{157.4}{9.7}\right)^2 \approx 263.31$

1.6 Lines

SLOPE-INTERCEPT FORM

$$y = mx + b,$$

where m is the **slope** (the number multiplying x) and b is the ***y*-intercept** (the number standing alone). This is the simplest model in mathematics, and in this course it is the model we will fit to data.

Exercise 18

For the following equations of lines, find the slope and the y-intercept.

(a) $y = 3x + 8$

(b) $y = 16x - 5.67$

(c) $y = -2 + 5x$

(d) $y = 0.00789 - 1.2345x$

Solution. Match each equation against $y = mx + b$. Watch the *sign* attached to each number, and do not be fooled by the order in which the terms are written.

	equation	slope m	intercept b
(a)	$y = 3x + 8$	3	8
(b)	$y = 16x - 5.67$	16	-5.67
(c)	$y = -2 + 5x$	5	-2
(d)	$y = 0.00789 - 1.2345x$	-1.2345	0.00789

In (c) and (d) the two terms are written in the opposite order. The slope is still whatever multiplies x, and the intercept is still the lone number — position on the page decides nothing.

1.7 Evaluating formulas

In this course the work is more about reasoning than about computation. Even so, you must be able to evaluate a new formula correctly on a basic scientific calculator. A good habit: first run the formula with easy numbers you can check in your head, so that you learn the right order of keystrokes. Only then put in the messy numbers.

Exercise 19

Let $l = x + f \cdot d$.

i. Find l when $f = 3,\ d = 2,\ x = 11$.

ii. Find l when $d = -8,\ f = 3,\ x = 11$.

iii. Find l when $x = 8.79,\ d = 1.7,\ f = 3.23$.

Solution. Multiplication happens before addition, so compute $f \cdot d$ first and only then add x.

i. $l = 11 + 3 \cdot 2 = 11 + 6 = 17.$

ii. $l = 11 + 3 \cdot (-8) = 11 - 24 = -13.$

iii. $l = 8.79 + 3.23 \cdot 1.7 = 8.79 + 5.491 = 14.281.$

In (ii) the product is negative, so adding it moves the answer down. If your calculator returned -104 you typed $(11 + 3) \cdot (-8)$: the parentheses were not there.

i. 17 ii. -13 iii. 14.281

Exercise 20

Let $w = \dfrac{x - m}{s/\sqrt{n}}$.

i. Find w when $x = 9,\ m = 6,\ s = 8,\ n = 16$.

ii. Find w when $x = 0.49,\ m = 0.5,\ s = 0.063,\ n = 17$.

Solution. Build the numerator and the denominator separately, then divide once at the end.

i. Numerator $9 - 6 = 3$. For the denominator, $\sqrt{16} = 4$, so $s/\sqrt{n} = 8/4 = 2$. Hence

$$w = \frac{3}{2} = 1.5.$$

ii. Numerator $0.49 - 0.5 = -0.01$, which is negative — that is fine and it means x sits below m. For the denominator, $\sqrt{17} = 4.12310\ldots$, so

$$\frac{s}{\sqrt{n}} = \frac{0.063}{4.12310\ldots} = 0.0152798\ldots$$

and therefore

$$w = \frac{-0.01}{0.0152798\ldots} = -0.65446\ldots \approx -0.654.$$

i. $w = 1.5$ ii. $w \approx -0.654$

CAREFUL

This w is not a random formula: it is the z-statistic we will use all term. Get the keystrokes right now. On most calculators you must write the denominator inside parentheses, `(0.063/\17)`, or the machine will divide only by 0.063 and then multiply by $\sqrt{17}$.

Exercise 21

Let $R = \sqrt{\dfrac{m_1^2}{n_1} + \dfrac{m_2^2}{n_2}}$.

i. Find R when $m_1 = 6,\ n_1 = 4,\ m_2 = 8,\ n_2 = 4$.

ii. Find R when $m_1 = 6.21,\ n_1 = 32,\ m_2 = 8.73,\ n_2 = 49$.

Solution. Everything under the root must be finished before the root is taken.

i.

$$\frac{6^2}{4} + \frac{8^2}{4} = \frac{36}{4} + \frac{64}{4} = 9 + 16 = 25, \qquad R = \sqrt{25} = 5.$$

ii.

$$\frac{6.21^2}{32} = \frac{38.5641}{32} = 1.205128\ldots, \qquad \frac{8.73^2}{49} = \frac{76.2129}{49} = 1.555365\ldots$$

Adding, $1.205128 + 1.555365 = 2.760493\ldots$, and then

$$R = \sqrt{2.760493\ldots} = 1.66147\ldots \approx 1.661.$$

i. $R = 5$ ii. $R \approx 1.661$

CAREFUL

A square root does *not* distribute over a sum:

$$\sqrt{x+y} \neq \sqrt{x} + \sqrt{y}.$$

In part (i), $\sqrt{9+16} = \sqrt{25} = 5$, whereas $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$. Different answers. Add first, then take the root.

Exercise 22

Let $r = \dfrac{m-t}{s}$.

i. Find m when $r = 2,\ s = 3,\ t = 4$.

ii. Find m when $r = 1.84,\ s = 3.12,\ t = 3.1$.

Solution. This time the unknown is not the subject of the formula, so rearrange first. Multiply both sides by s and then move t across:

$$r = \frac{m-t}{s} \implies rs = m - t \implies m = rs + t.$$

Only now substitute.

i. $m = 2 \cdot 3 + 4 = 6 + 4 = 10.$

ii. $m = 1.84 \cdot 3.12 + 3.1 = 5.7408 + 3.1 = 8.8408 \approx 8.841.$

i. $m = 10$ ii. $m \approx 8.841$

1.8 Summation notation

The notation $\sum_{i=1}^{n} x_i$ stands for the sum of n numbers $x_1, x_2, \ldots, x_n$, and is read *the sum of all* x_i *with* i *running from* 1 *to* n:

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + x_3 + \cdots + x_n.$$

The term following the sign $\sum$ (the Greek letter sigma) says *what* is being added; the labels below and above say *how many* terms there are.

Exercise 23

Let x represent the number of M&M's contained in individual trail mix snacks. The data for five trail mix snacks is recorded below.

Trail mix (i)	1	2	3	4	5
M&M's count (x_i)	6	5	6	7	8

(a) The formula that means to sum up the M&M's counts of all five trail mixes is $\sum_{i=1}^{5} x_i$. Evaluate it.

(b) Show the formula that means to sum up the M&M's counts of the first three trail mixes, and evaluate it.

(c) The formula that means to sum up the squares of all M&M's counts is $\sum_{i=1}^{5} x_i^2$. Evaluate it.

(d) The formula for the square of the sum of all M&M's counts is $\left(\sum_{i=1}^{5} x_i\right)^2$. Evaluate it.

(e) Fill in the blank with the proper symbol, $=$ or $\neq$: $\displaystyle\sum_{i=1}^{5} x_i^2 __ \left(\sum_{i=1}^{5} x_i\right)^2$.

Solution. **(a)** Add the five counts:

$$\sum_{i=1}^{5} x_i = x_1 + x_2 + x_3 + x_4 + x_5 = 6 + 5 + 6 + 7 + 8 = 32.$$

(b) "The first three trail mixes" means stopping at $i = 3$. What is being added is still x_i, the counts — not the labels i:

$$\sum_{i=1}^{3} x_i = x_1 + x_2 + x_3 = 6 + 5 + 6 = 17.$$

(c) Now each count is squared *before* adding:

$$\sum_{i=1}^{5} x_i^2 = 6^2 + 5^2 + 6^2 + 7^2 + 8^2 = 36 + 25 + 36 + 49 + 64 = 210.$$

(d) Here the adding happens first and the squaring last. Using the total from (a),

$$\left(\sum_{i=1}^{5} x_i\right)^2 = 32^2 = 1024.$$

(e) Comparing (c) and (d), 210 against 1024:

$$\sum_{i=1}^{5} x_i^2 \neq \left(\sum_{i=1}^{5} x_i\right)^2.$$

(a) 32 (b) 17 (c) 210 (d) 1024 (e) $\neq$

CAREFUL

Parts (c) and (d) look almost the same written down, and they are not the same number at all: the sum of the squares is 210, the square of the sum is 1024. Square-then-add and add-then-square are different instructions. Both appear in the formula for the standard deviation, so the distinction will cost you marks if you let it slide now.

CHAPTER 2

Displaying and Describing Data

2.1 What statistics is about

Three ideas run underneath everything we will do this term.

> **THE THREE IDEAS**
>
> **Variation.** Data vary, because we do not see everything, and even what we do see we measure imperfectly.
> **Learning from the data.** We hope to learn about the world as well as we can from the limited, imperfect data we have.
> **Making intelligent decisions.** The better we understand the world, the wiser our decisions will be.

Notice that the first idea is not an apology. Variation is not a defect in the data to be removed; it is the subject matter. Statistics is the discipline of saying something reliable in spite of it, and of being honest about how reliable.

2.2 The *Who* and the *What*

In general the rows of a data table correspond to individual **cases**, about whom — or about which, if they are not people — we record some characteristics. Cases go by different names depending on the situation:

- individuals who answer a survey are **respondents**;
- people on whom we experiment are **subjects**, or **participants**;
- animals, plants, websites and other inanimate subjects are often called **experimental units**;
- in a database the rows are **records**;
- often we simply call cases what they are: customers, economic quarters, companies.

Whatever we call them, the rows are the *Who* of the data.

The characteristics recorded about each individual are the **variables**. They are usually the columns of the data table, with a name in the header saying what has been recorded. In a table of music orders, the variables are the order number, the name, the state or country, the price, the area code, and so on.

Categorical and quantitative variables

Some variables just tell us what group or category each individual belongs to. Are you male or female? Pierced or not? We call these **categorical** (or qualitative) variables. Their values

are often text — the variable *state/country* is clearly categorical — but *numerals are often used as labels*, so a categorical variable can perfectly well look numeric. An area code such as 13905 in Binghamton is a categorical variable: no one adds or averages area codes.

When a variable contains measured numerical values with units, we call it a **quantitative** variable.

Exercise 1 (Consumer Reports tablets)

A *Consumer Reports* article about 126 tablets lists each tablet's manufacturer, price, battery life (hrs.), the operating system (Android, iOS or Windows), an overall quality score (0–100), and whether or not it has a memory card reader. Are these variables categorical or quantitative? Include units where appropriate, and describe the *Why* of this investigation.

Solution. Ask of each one: does it name a group, or does it measure an amount with units?

variable	type	units
manufacturer	categorical	—
price	quantitative	dollars
battery life	quantitative	hours
operating system	categorical	—
quality score	quantitative	none (a 0–100 scale)
memory card reader	categorical	—

The *Who* is the sample of 126 tablets; each one selected represents all similar tablets offered by that manufacturer. The *Why* is that the magazine hopes to give consumers the information they need to choose a good tablet.

CAREFUL

The quality score is quantitative even though it has no units, because it measures an amount on a scale where larger means more. The operating system is categorical even if someone codes Android as 1, iOS as 2 and Windows as 3. Type is decided by what the variable *means*, never by whether it looks like a number.

2.3 How the data are collected

How the data are collected can make the difference between insight and nonsense. We will require, from here on, that the data be collected **randomly** — or at least that we are willing to assume so. Later we will study that random condition properly, with probability and with models for distributions.

CAREFUL

The random condition is not a formality. Without it we can reach confident, well-computed, completely wrong conclusions. A survey of who likes a colour of shirt, taken among people already wearing it, produces excellent arithmetic about nothing.

2.4 Models

A **model** is a summary and a simplification of the data, which helps our understanding in many ways. It is a simplification of reality that gives us information we can learn from and use, even though it does not represent reality exactly.

Kepler's laws are the great example. From Tycho Brahe's observations of the declination of Mars over the twenty years before 1600 — a cloud of points with gaps in it — Kepler saw through the small anomalies of measurement and produced three simple laws. The value of the model is visible exactly where the data are not: for a year such as 1598 there are no observations, and yet the curve gives us a declination of about 25 degrees for it. With a model we obtain a great deal of information about our data, and we can also ask about the future.

2.5 Frequency tables and bar charts

Categorical variables are easy to summarize in a **frequency table**, which lists the number of cases in each category along with its name. A **relative frequency table** displays percentages, or proportions, instead of counts. Both show the **distribution** of a categorical variable, because both name the possible categories and say how often each one occurs.

> **Exercise 2** (Titanic ticket class)
>
> Among the people aboard the *Titanic*, the categorical variable *ticket class* had these counts: first class 324, second class 285, third class 710, crew 889. Build the frequency table and the relative frequency table.

Solution. The counts are given, so the frequency table is immediate. For the relative frequency table, every percentage is $\frac{\#1}{\text{total}}$, and the total must be built first:

$$\text{total} = 324 + 285 + 710 + 889 = 2208.$$

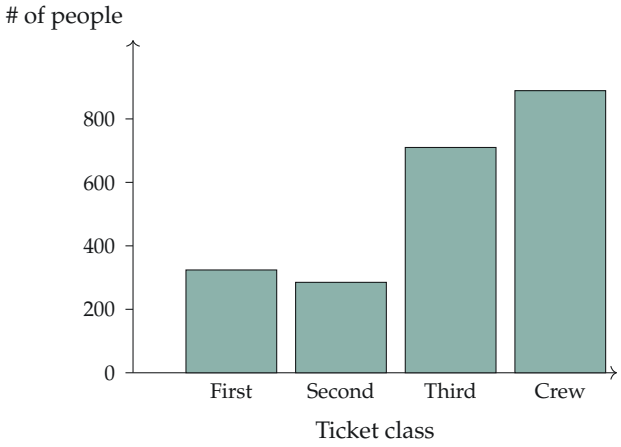
Then, one category at a time,

$$\frac{324}{2208} = 0.1467\ldots = 14.67\%, \qquad \frac{285}{2208} = 12.91\%,$$
$$\frac{710}{2208} = 32.16\%, \qquad \frac{889}{2208} = 40.26\%.$$

class	count	percentage
First	324	14.67%
Second	285	12.91%
Third	710	32.16%
Crew	889	40.26%
Total	2208	100.00%

Here the percentages add to exactly 100%, but they need not: rounding each one separately can leave the sum a little above or below 100.

A **bar chart** displays the same distribution as bars, one per category, with gaps between them, because the categories have no order of their own.



Now it is easy to see that the majority of the people on board were not crew.

2.6 Histograms

We cannot make a bar chart for a quantitative variable, because quantitative variables have no categories. Instead we make a **histogram**.

> **BAR CHART OR HISTOGRAM**
>
> Both use bars, and they are fundamentally different.
> In a **bar chart** the horizontal axis just names categories. The bars could be put in any order, and they are drawn with a space between them.
> In a **histogram** the horizontal axis shows the values of the variable *in order*. The axis is sliced into equal-width **bins**, and the bars show the count in each bin. Here the bars touch, and a **gap** is meaningful: it marks a region where there are no observations at all.

In the histogram of the ages of those aboard the *Titanic*, each bin is 5 years wide, so the tallest bar says that the most populous age group was the 20 to 24 year olds, with over 400 people. The youngest passengers were infants and the oldest was more than 70 years old.

2.7 Shape

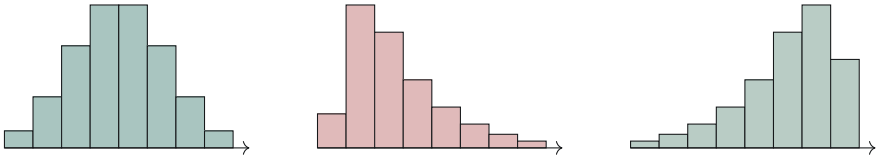
We summarize the shape of a distribution in terms of three properties: how many modes it has, whether it is symmetric or skewed, and whether it has any extraordinary cases, or outliers.

Modes

A hump in the histogram is called a **mode**. For a categorical variable the mode is simply the most frequent category; for a quantitative variable it makes more sense to use the word for the *peak* of the histogram than for a single value. One peak is **unimodal**, two peaks **bimodal**, three or more **multimodal**. A histogram with no mode at all, in which the bars are all about the same height, is called **uniform**.

Symmetry and skewness

A distribution is **symmetric** if you could fold the histogram along a vertical line through the middle and have the edges match closely. The thinner ends of a distribution are its **tails**, and if one tail stretches out farther than the other the histogram is said to be **skewed** to the side of the longer tail.



symmetric skewed to the right skewed to the left

We generally prefer symmetric distributions, because they are easier to summarize, to model and to discuss: it is pretty clear where the centre is.

Outliers and gaps

Always mention any stragglers, or **outliers**, that stand away from the body of the distribution. If you were collecting data on nose lengths and Pinocchio were in the group, he would be an outlier, and you would certainly want to mention him.

> **WHAT TO DO WITH AN OUTLIER**
>
> An outlier can be the most informative part of your data. Or it might just be an error. *Never throw data away without comment.* Treat outliers specially and discuss them when you tell about your data; or find the error and fix it if you can. Be sure to look for outliers — always.

You should also notice any **gaps**, where no data appear at all. Gaps may indicate separate modes, or even separate subgroups worth considering individually. But do not be overzealous: in a small data set a gap may be pure happenstance.

> **Exercise 3** (Credit card expenditures)
>
> A credit card company provides data on the amount spent by 500 selected customers during a three-month period. The histogram shows a very tall bar near zero, a long thin tail to the right reaching an extraordinarily large value at about $7000, and a few values below zero. Describe the shape of this distribution.

Solution. Answer the three questions of shape, in order.

Modes: one tall peak at the low end, so the distribution is unimodal.

Symmetry: the right tail is very much longer than the left, so it is skewed to the high end — skewed to the right.

Unusual features: there is an extraordinarily large value at about $7000, which should be mentioned and investigated rather than deleted. Some of the expenditures are negative, which for a credit card is not an error at all: a refund or a credit produces a negative charge.

2.8 The centre: median and mean

The median

One natural choice of typical value is the **median**: the value literally in the middle, with half the values below it and half above. The median has the same units as the data — say so whenever you report one. The median age of crew members on the *Titanic* was 30 years.

FINDING THE MEDIAN BY HAND

Order the values first; everything else depends on it.

If n is **odd**, the median is the single middle value, in position $\dfrac{n+1}{2}$.

If n is **even**, there are two middle values, in positions $\dfrac{n}{2}$ and $\dfrac{n}{2}+1$, and the median is their average.

Exercise 4 (Median by hand)

Find the median of

$$14.1, \quad 3.2, \quad 25.3, \quad 2.8, \quad -17.5, \quad 13.9, \quad 45.8,$$

and then of the same batch with one more value, 35.7, added.

Solution. **First batch.** Order the values:

$$-17.5,\ 2.8,\ 3.2,\ \mathbf{13.9},\ 14.1,\ 25.3,\ 45.8.$$

There are $n = 7$ values, an odd number, so the median sits in position $\frac{7+1}{2} = 4$. Counting in from either end, that is 13.9. Three values are lower and three are higher, as it should be.

Second batch. With 35.7 added the ordered values are

$$-17.5,\ 2.8,\ 3.2,\ \mathbf{13.9},\ \mathbf{14.1},\ 25.3,\ 35.7,\ 45.8.$$

Now $n = 8$ is even, so we average the values in positions $\frac{8}{2} = 4$ and $\frac{8}{2}+1 = 5$:

$$\text{median} = \frac{13.9 + 14.1}{2} = 14.0.$$

Four data values are lower and four are higher.

13.9 and 14.0

The mean

The most common way to summarize the centre of a distribution is with the **mean**:

$$\bar{y} = \frac{\text{total}}{n} = \frac{\sum y_i}{n},$$

where n is the sample size, that is, the number of data values.

The mean is located at the *balancing point* of the histogram — of the histogram, not of a bar chart. Picture the histogram cut out of stiff card: the mean is where it would balance on a knife edge.

Which one, the mean or the median?

Both summarize the centre, and they do it in different ways. The median is *less influenced by skewness or by outliers* than the mean, which makes it a good choice for skewed distributions such as income, or company size, or the price of a ticket.

The *Titanic* fares show this clearly. The distribution of ticket prices is skewed to the right; the mean is £33.0 while the median is £14.4. The handful of very expensive tickets at the right have pulled the mean towards them and away from the median. The mean is not the middle point.

CHOOSING A CENTRE

For a **skewed** distribution, report the **median**.
For a **symmetric** distribution, the **mean** is the better summary.
When the two differ a lot, that difference is itself information: it is telling you the distribution is skewed.

2.9 The spread

We also want to measure the **spread**: how much the data values vary around the centre. When you describe a distribution numerically, always report a measure of its spread along with its centre. Spread, like shape and centre, is a general description of the distribution, not a precise quantity.

The range

$$\text{Range} = \max - \min\,.$$

The range is a single number, not an interval, despite the way the word is used in ordinary speech. The maximum *Titanic* crew age was 62 and the minimum was 14 — the bellboy — so the range is $62 - 14 = 48$ years.

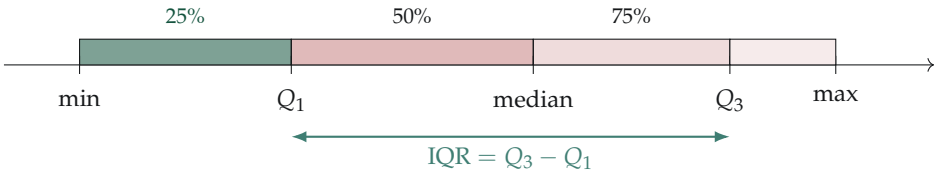
Its weakness is that one extreme value can make it enormous and unrepresentative, so the range should never be your only measure of spread.

Quartiles and the interquartile range

The value just above the lower 25% of the data is the **lower quartile**, or first quartile, or 25th percentile, written Q_1. The **upper quartile** Q_3 is the 75th percentile. The median is the 50th percentile. The distance between the quartiles is the **interquartile range**:

$$\text{IQR} = Q_3 - Q_1,$$

the range of just the middle half of the data.



For the *Titanic* crew ages the median is 30 years: ordering the 889 ages, the 445th is 30. Taking the lower 445 ages, their median is 24, and that is Q_1; in the upper half the median is 37, and that is Q_3. So

$$\text{IQR} = 37 - 24 = 13 \text{ years}.$$

The range is 48 years and the middle half covers only 13: that contrast is the useful part.

> **Exercise 5** (Spread of the credit card expenditures)
>
> The distribution of credit card expenditures is skewed, and the quartiles are \$73.84 and \$624.80. What is the IQR, and why is it a suitable measure of spread here?

Solution.

$$\text{IQR} = 624.80 - 73.84 = 550.96 \text{ dollars}.$$

It is suitable for the same reason the median was: like the median, the IQR is built from positions in the ordered data rather than from the values themselves, so it is not affected by the outlying \$7000 charge or by the skewness. The range, by contrast, would be dominated by that single customer.

IQR = \$550.96

2.10 Variance and standard deviation

The IQR is almost always a reasonable summary, but because it uses only the two quartiles it ignores much of the information about how the individual values vary. The **standard deviation** takes into account how far *each* value is from the mean.

We first define the **variance**:

$$s^2 = \frac{\sum (y_i - \bar{y})^2}{n - 1},$$

where $\bar{y}$ is the mean, n the sample size and y_i each data value.

> **Exercise 6** (Variance and standard deviation by hand)
>
> For the data $1,\ 3,\ 7,\ 22,\ 55$, compute the mean, the variance and the standard deviation.

Solution. Here $n = 5$ and $y_1 = 1$, $y_2 = 3$, $y_3 = 7$, $y_4 = 22$, $y_5 = 55$.

The mean.

$$\bar{y} = \frac{\sum y_i}{n} = \frac{1 + 3 + 7 + 22 + 55}{5} = \frac{88}{5} = 17.6.$$

The deviations and their squares. Subtract the mean from each value and square; it is worth laying this out in a table, because that is where the arithmetic goes wrong.

y_i	$y_i - \bar{y}$	$(y_i - \bar{y})^2$
1	-16.6	275.56
3	-14.6	213.16
7	-10.6	112.36
22	4.4	19.36
55	37.4	1398.76
		2019.20

The variance. Divide by $n - 1 = 4$, not by n:

$$s^2 = \frac{2019.20}{4} = 504.80.$$

The standard deviation. The variance is in squared units, which is why we take the square root:

$$s = \sqrt{s^2} = \sqrt{504.80} = 22.467\ldots \approx 22.47.$$

$\bar{y} = 17.6, \quad s^2 = 504.80, \quad s \approx 22.47$

CAREFUL

Two places to be careful. First, the denominator is $n - 1$, not n: dividing by 5 here would give 403.84, which is not the sample variance. Second, the deviations must be squared *before* they are added — they sum to zero otherwise, every time, which is exactly why we square them.

UNITS

The variance is in squared units: squared dollars, squared years, mpg^2. Nobody wants to discuss squared ages, and that is the whole reason the standard deviation exists:

$$s = \sqrt{\frac{\sum(y - \bar{y})^2}{n - 1}}$$

is back in the units of the data. In practice we almost always let technology compute s, but you should be able to do it by hand once, as above, so that you know what the machine is doing.

2.11 Putting it together

Exercise 7 (Question for you: the five-number summary)

How do we compute Q_1, the median and Q_3 from a set of data? Do it for 1, 3, 7, 22, 55, and give the range and the IQR as well.

Solution. **Step 1: order the data.** They already are: 1, 3, 7, 22, 55.

Step 2: the median. Here $n = 5$ is odd, so the median is the value in position $\frac{5+1}{2} = 3$, that is

$$\text{median} = 7.$$

Step 3: the quartiles. Q_1 is the median of the lower half and Q_3 the median of the upper half. With an odd n the median itself is *not* included in either half, so

$$\text{lower half} = \{1,\ 3\} \implies Q_1 = \frac{1+3}{2} = 2,$$
$$\text{upper half} = \{22,\ 55\} \implies Q_3 = \frac{22+55}{2} = 38.5.$$

Step 4: range and IQR.

$$\text{Range} = 55 - 1 = 54, \qquad \text{IQR} = 38.5 - 2 = 36.5.$$

The five-number summary is therefore

$$\min = 1, \quad Q_1 = 2, \quad \text{median} = 7, \quad Q_3 = 38.5, \quad \max = 55.$$

Compare this with the mean, 17.6, computed in the previous exercise. The mean is more than twice the median, which tells us at once — without drawing anything — that this little batch is strongly skewed to the right, pulled by the 55.

$Q_1 = 2, \quad \text{median} = 7, \quad Q_3 = 38.5, \quad \text{IQR} = 36.5$

THE REPORTING HABIT

Whenever you describe a distribution, say three things and in this order: **shape**, then **centre**, then **spread** — and let the shape decide the other two. Skewed: median and IQR. Symmetric: mean and standard deviation. Then mention any outliers separately, by name.